

CSE4067 - Introduction to Blockchain Programming

Fall 2024-2025, Final Exam

Student No:

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Problem 1 (+15)

Explain *Proof-of-Work (PoW)* and *Proof-of-Stake (PoS)* consensus mechanisms and highlight their main differences.

Problem 2 (+20)

Suppose you are given a file that contains the following bits (given in hexadecimal):

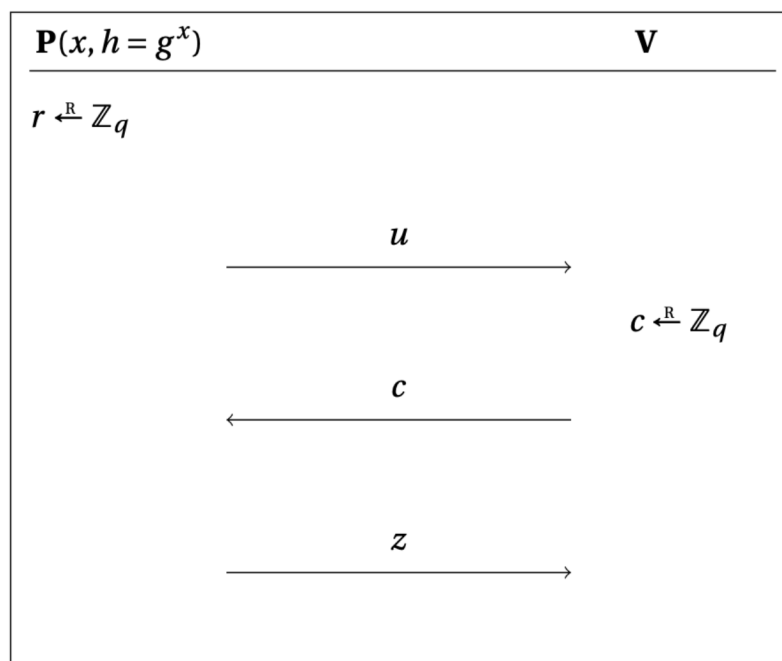
9f47c13d8e2af6b5

Divide the file into data chunks of *16 bits* and construct the Merkle tree. Take the hash function to be the *8-bit* block-wise *XNOR* function.

Problem 3 (+30)

Simulate the following zero-knowledge proof with *Schnorr's Protocol* between Prover (P) and Verifier (V) where (secret value $x = 3$), (public modulus $n = 17$) and (generator $g = 2$).

- Show all calculations both for $c = 0$ and $c = 1$ including how verifier verifies the proofs. Follow the *sigma* protocol template below for interactions.
- Explain why privacy of secret value is eventually satisfied.
- Explain how to transform it to a *non-interactive* proof protocol.



Problem 4 (+35)

Write the following *Solidity* smart contract that will implement an *autonomous mobile theft insurance company service (InsuranceDAO)*:

- When a person buys insurance, his/her phone is insured against thefts after the payment. Each address can only be used for one phone.
- The cost of insurance is 1 Ether.
- When a phone is stolen, the person reports the incident with the purchase no of the phone to the police. If the police approves the report, it records the address and the cost of the phone (in *wei*) in the *StolenPhones* contract.
- The information about whether the phone is stolen can be queried from the police by calling *isItStolen()* function in the *StolenPhones* contract. It returns the cost of the phone, if the phone is stolen or 0 otherwise.
- If there is enough balance in the *InsuranceDAO* contract, insurance equal to the cost of the phone (obtained from the *StolenPhones* contract) is paid by calling *getCompensation()* function.
- If the money in the *InsuranceDAO* contract is less than the cost of the phone, then all the money left is paid. If there is no money left at all, nothing is paid.